

```

GNU nano 8.0 /etc/ipsec.d/tunnel-to-branchB.conf
conn branchA-to-branchB
    type=tunnel
    left=192.168.3.168
    leftsubnet=10.0.1.0/24
    leftid=192.168.3.168
    right=192.168.3.194
    rightsubnet=10.0.2.0/24
    rightid=192.168.3.194
    authby=secret
    auto=start
    ike=aes256-sha2-modp2048
    esp=aes256-sha2
    leftsendcert=never
    rightsendcert=never

```

```

GNU nano 8.0 /etc/ipsec.d/tunnel-to-branchA.conf
conn branchB-to-branchA
    type=tunnel
    # Наша сторона (Branch B)
    left=192.168.3.194
    leftsubnet=10.0.2.0/24
    leftid=192.168.3.194

    # Сторона партнера (Branch A)
    right=192.168.3.168
    rightsubnet=10.0.1.0/24
    rightid=192.168.3.168

    authby=secret
    auto=start
    ike=aes256-sha2-modp2048
    esp=aes256-sha2
    leftsendcert=never
    rightsendcert=never

```

```

[root@localhost ~]# ipsec status
000 using kernel interface: xfrm
000
000 interface lo UDP [::1]:4500
000 interface lo UDP [::1]:500
000 interface lo UDP 127.0.0.1:4500
000 interface lo UDP 127.0.0.1:500
000 interface ens33 UDP 192.168.3.194:4500
000 interface ens33 UDP 192.168.3.194:500
000 interface ens37 UDP 10.0.2.1:4500
000 interface ens37 UDP 10.0.2.1:500
000
000 fips mode=disabled;
000 SELinux=enabled
000 seccomp=disabled
000
000 config setup options:
000
000 configdir=/etc, configfile=/etc/ipsec.conf, secrets=/etc/ipsec.secrets, ipsecdir=/etc/ipsec.d
000 nssdir=/var/lib/ipsec/nss, dumpdir=/run/pluto, statsbin=unset
000 dnssec-rootkey-file=/var/lib/unbound/root.key, dnssec-trusted=<unset>
000 sbindir=/usr/sbin, libexecdir=/usr/libexec/ipsec
000 pluto version=4.15, pluto_vendorid=0E-Libreswan-4.15, audit-log=yes
000 nhelpers=-1, uniqueids=yes, dnssec-enable=yes, logappend=yes, logip=yes, shuntlifetime=900s, xfrmlifetime=30s
000 ddos-cookies-threshold=25000, ddos-max-halfopen=50000, ddos-mode=auto, ikev1-policy=accept
000 ikebuf=0, msg_errqueue=yes, crl-strict=no, crlcheckinterval=0, listen=<any>, nlog-all=0
000 ocsdp-enable=no, ocsdp-strict=no, ocsdp-timeout=2, ocsdp-uri=<unset>
000 ocsdp-trust-name=<unset>
000 ocsdp-cache-size=1000, ocsdp-cache-min-age=3600, ocsdp-cache-max-age=86400, ocsdp-method=get
000 global-redirect=no, global-redirect-to=<unset>
000 secctx-attr-type=32001
000 debug:
000
000 nat-traversal=yes, keep-alive=20, nat-ikeport=4500
000 virtual-private (%priv):
000
000 Kernel algorithms supported:
000
000 algorithm ESP encrypt: name=3DES_CBC, keysize=192, keysize=192
000 algorithm ESP encrypt: name=AES_CBC, keysize=128, keysize=256
000 algorithm ESP encrypt: name=AES_CCM_12, keysize=128, keysize=256

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000
000 stats db_ops: {curr_cnt, total_cnt, maxsz} :context={0,0,0} trans={0,0,0} attrs={0,0,0}
000
000 Connection list:
000
000 "branchB-to-branchA": 10.0.2.0/24===192.168.3.194[+S-C]...192.168.3.168[+S-C]===10.0.1.0/24; erouted; eroute owner: #3
000 "branchB-to-branchA": oriented; my_ip=unset; their_ip=unset; my_updown=ipsec_updown;
000 "branchB-to-branchA": xauth us:none, xauth them:none, my_username=[any]; their_username=[any]
000 "branchB-to-branchA": our_auth:secret, their_auth:secret, our_authheap:none, their_authheap:none;
000 "branchB-to-branchA": modecfg_info: us:none, them:none, modecfg_policy:push, dns:unset, domains:unset, cat:unset;
000 "branchB-to-branchA": sec_label:unset;
000 "branchB-to-branchA": ike_life: 28800s; ipsec_life: 28800s; ipsec_max_bytes: 2^63B; ipsec_max_packets: 2^63; replay_window: 128; rekey_margin: 540s; r
ekey_fuzz: 100%; keyingtries: 0;
000 "branchB-to-branchA": retransmit-interval: 500ms; retransmit-timeout: 60s; iketcp:no; iketcp-port:4500;
000 "branchB-to-branchA": initial-contact:no; cisco-unity:no; fake-strongswan:no; send-vendorid:no; send-no-esp-tfc:no;
000 "branchB-to-branchA": policy: IKEv2+PSK+ENCRYPT+TUNNEL+PFS+UP+IKE_FRAG_ALLOW+ESN_NO+ESN_YES;
000 "branchB-to-branchA": v2-auth-hash-policy: none;
000 "branchB-to-branchA": conn_prio: 24,24; interface: ens33; metric: 0; mtu: unset; sa_prio:auto; sa_tfc:none;
000 "branchB-to-branchA": nflog-group: unset; mark: unset; vti-iface:unset; vti-routing:no; vti-shared:no; nic-offload:auto;
000 "branchB-to-branchA": our_idtype: ID_IPV4_ADDR; our_id=192.168.3.194; their_idtype: ID_IPV4_ADDR; their_id=192.168.3.168
000 "branchB-to-branchA": liveness: passive; dpdaction:hold; dpddelay:0s; retransmit-timeout:60s
000 "branchB-to-branchA": nat-traversal: encaps:auto; keepalive:20s
000 "branchB-to-branchA": newest IKE SA: #1; newest IPsec SA: #3; conn serial: $1;
000 "branchB-to-branchA": IKE algorithms: AES_CBC_256-HMAC_SHA2_256-MODP2048
000 "branchB-to-branchA": IKEv2 algorithm newest: AES_CBC_256-HMAC_SHA2_256-MODP2048
000 "branchB-to-branchA": ESP algorithms: AES_CBC_256-HMAC_SHA2_256_128
000 "branchB-to-branchA": ESP algorithm newest: AES_CBC_256-HMAC_SHA2_256_128; pfsgroup=<Phase1>
000
000 Total IPsec connections: loaded 1, active 1
000
000 State Information: DoS cookies not required, Accepting new IKE connections
000 IKE SAs: total(1), half-open(0), open(0), authenticated(1), anonymous(0)
000 IPsec SAs: total(2), authenticated(2), anonymous(0)
000
000 #1: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_IKE_SA (established IKE SA); REKEY in 28040s; REPLACE in 28769s; newest; idle;
000 #2: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 28165s; REPLACE in 28769s; IKE SA #1; idle;
000 #2: "branchB-to-branchA" esp.82de6d1b@192.168.3.168 esp.5e873f7f@192.168.3.194 tun.0@192.168.3.168 tun.0@192.168.3.194 Traffic: ESPin=0B ESPout=0B ESPma
x=2^63B
000 #3: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 27904s; REPLACE in 28784s; newest; eroute owner; IKE SA #1;
idle;
000 #3: "branchB-to-branchA" esp.715e3a62@192.168.3.168 esp.399639f6@192.168.3.194 tun.0@192.168.3.168 tun.0@192.168.3.194 Traffic: ESPin=0B ESPout=0B ESPma
x=2^63B
000
000 Bare Shunt list:
000
[root@localhost ~]# |

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000
[root@localhost ~]# ping -c 4 -I 10.0.2.1 10.0.1.1
PING 10.0.1.1 (10.0.1.1) from 10.0.2.1 : 56(84) bytes of data.
64 bytes from 10.0.1.1: icmp_seq=1 ttl=64 time=1.85 ms
64 bytes from 10.0.1.1: icmp_seq=2 ttl=64 time=0.755 ms
64 bytes from 10.0.1.1: icmp_seq=3 ttl=64 time=0.968 ms
64 bytes from 10.0.1.1: icmp_seq=4 ttl=64 time=0.413 ms

--- 10.0.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.413/0.996/1.850/0.531 ms
[root@localhost ~]# |

```

```
root@localhost:~# ipsec status | grep ESTABLISHED
000 Total IPsec connections: loaded 1, active 1
000 State Information: DoS cookies not required, Accepting new IKE connections
000 IKE SAs: total(1), half-open(0), open(0), authenticated(1), anonymous(0)
000 IPsec SAs: total(2), authenticated(2), anonymous(0)
000 #66: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_IKE_SA (established IKE SA); REKEY in 28454s; REPLACE in 28724s; newest; idle;
000 #67: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 28454s; REPLACE in 28724s; IKE SA #66; idle;
000 #67: "branchA-to-branchB" esp.5e873f7f@192.168.3.194 esp.82de6d1b@192.168.3.168 tun.0@192.168.3.194 tun.0@192.168.3.168 Traffic: ESPin=0B ESPout=0B ESPmax=2*63B
000 #68: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 28470s; REPLACE in 28740s; newest; eroute owner; IKE SA #66; idle;
000 #68: "branchA-to-branchB" esp.399639f6@192.168.3.194 esp.715e3a62@192.168.3.168 tun.0@192.168.3.194 tun.0@192.168.3.168 Traffic: ESPin=0B ESPout=0B ESPmax=2*63B
000 Bare Shunt list:
000
[root@localhost ~]# ipsec status | grep ESTABLISHED
000 #66: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_IKE_SA (established IKE SA); REKEY in 28213s; REPLACE in 28483s; newest; idle;
000 #67: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 28213s; REPLACE in 28483s; IKE SA #66; idle;
000 #68: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 28229s; REPLACE in 28499s; IKE SA #66; idle;
000 #69: "branchA-to-branchB":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 27971s; REPLACE in 28559s; newest; eroute owner; IKE SA #66; idle;
[root@localhost ~]#

000 #2: "branchB-to-branchA" esp.82de6d1b@192.168.3.168 esp.5e873f7f@192.168.3.194 tun.0@192.168.3.168 tun.0@192.168.3.194 Traffic: ESPin=0B ESPout=0B ESPmax=2*63B
000 #3: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 27904s; REPLACE in 28784s; newest; eroute owner; IKE SA #1; idle;
000 #3: "branchB-to-branchA" esp.715e3a62@192.168.3.168 esp.399639f6@192.168.3.194 tun.0@192.168.3.168 tun.0@192.168.3.194 Traffic: ESPin=0B ESPout=0B ESPmax=2*63B
000 Bare Shunt list:
000
[root@localhost ~]# ping -c 4 -I 10.0.2.1 10.0.1.1
PING 10.0.1.1 (10.0.1.1) from 10.0.2.1 : 56(84) bytes of data.
64 bytes from 10.0.1.1: icmp_seq=1 ttl=64 time=1.85 ms
64 bytes from 10.0.1.1: icmp_seq=2 ttl=64 time=0.755 ms
64 bytes from 10.0.1.1: icmp_seq=3 ttl=64 time=0.968 ms
64 bytes from 10.0.1.1: icmp_seq=4 ttl=64 time=0.413 ms

--- 10.0.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.413/0.996/1.850/0.531 ms
[root@localhost ~]# ipsec status | grep ESTABLISHED
000 #1: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_IKE_SA (established IKE SA); REKEY in 27760s; REPLACE in 28489s; newest; idle;
000 #2: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 27885s; REPLACE in 28489s; IKE SA #1; idle;
000 #3: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 27625s; REPLACE in 28505s; IKE SA #1; idle;
000 #4: "branchB-to-branchA":500 STATE_V2_ESTABLISHED_CHILD_SA (established Child SA); REKEY in 28295s; REPLACE in 28565s; newest; eroute owner; IKE SA #1; idle;
[root@localhost ~]#
```

После отключения:

```
[root@localhost ~]# ping -c 4 -I 10.0.2.1 10.0.1.1
ping: bind: Невозможно назначить запрошенный адрес
[root@localhost ~]# ping 10.0.2.1 10.0.1.1
PING 10.0.1.1 (10.0.1.1) 56(124) bytes of data.
^C
--- 10.0.1.1 ping statistics ---
17 packets transmitted, 0 received, 100% packet loss, time 16395ms
```